CONCEPT MAP · GENERATIVE AI

A Comparative Concept Map under Artificial Intelligence

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Course: CSA6503 – Generative AI and Large Scale Models

Artificial Intelligence

Includes ↓

Machine Learning

Contains ↓

Deep Learning

Machine Learning

DEFINITION

Algorithms learn patterns from data.

REQUIRES

Feature Engineering: Manual · **Training Data:** Moderate

COMPUTATION

Low to Medium

TRAINING TIME

Faster

EXAMPLES

Spam Detection, Recommendation Systems

💻 **ALGORITHMS**

Decision Tree · Random Forest · SVM · KNN

APPLICATIONS

Fraud Detection · Email Filtering · Sales Prediction

VS

- Needs fewer data ⇄ Needs large data
- Manual ⇄ Automatic feature extraction
- Lower ⇄ Higher computational cost
- Traditional algorithms ⇄ Neural networks ⚡

🧠 Deep Learning

DEFINITION

Uses multi-layer neural networks to learn automatically.

REQUIRES

Feature Engineering: Automatic · **Training Data:** Large Datasets

COMPUTATION

Very High (GPU/TPU)

TRAINING TIME

Longer

EXAMPLES

Face Recognition, ChatGPT, Self-driving Cars

⚡ **MODELS**

CNN · RNN · Transformer

APPLICATIONS

Image Recognition · Speech Recognition · Language Translation · Generative AI

📈 *Both branches share the same source of data — but differ in how deeply they learn from it.*

CONCLUSION

"Deep Learning is a specialized subset of Machine Learning that automatically learns complex patterns from large amounts of data."

REAL-WORLD EXAMPLES

📊 **Machine Learning**

- Netflix Recommendation
- Email Spam Filter

🧠 **Deep Learning**

- ChatGPT
- Google Translate
- Tesla Autopilot

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